

Many-body Lyapunov spectroscopy from maximally mixed Gaussian trajectories: Diagnostic results for the adaptive chiral wall

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We reconstruct the many-body singular spectrum of a resolved Gaussian trajectory from a maximally mixed input. The normalized conditional output is the right Choi marginal, $\hat{V}_m \hat{V}_m^\dagger / Z_m$; its one-particle occupations generate the normalized many-body spectrum, while the accumulated Born probability restores Z_m . We apply this construction to 800 independent Born trajectories at $N_x = 20$, eight circumferences $N_y = 20\text{--}40$, and depth $T = 2N_y$. All 160 task products pass checksum, provenance, reconstruction, cycle-completeness, exact-cap, and numerical-residual checks. Late-window record-weight slopes agree with the leading spectral slopes to better than 0.3% at every size, and the first four soft modes carry 95–97% of their transverse weight near the two walls. However, every circumference fails at least one preregistered middle-to-late stability test. The leading $1/N_y^2$ coefficient has the wrong sign, a confidence interval containing zero, and extreme fit-window sensitivity. The campaign therefore validates the Gaussian spectrum reconstruction and its boundary localization, but it does not support a precision effective-central-charge or absolute operator-dimension extraction. We recommend, but do not launch, a fresh $T = 4N_y$ depth study at $N_y = 20, 30, 40$.

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(mEO) $\hat{V}_m(t)$. Zabalo *et al.* call the corresponding trajectory operator K_m ; we retain $\hat{V}_m$ for the project's canonical mEO notation and use their notation for its singular values and Lyapunov exponents. We want the leading many-body singular values and their long-time Lyapunov rates without constructing an exponentially large Fock-space matrix.

The term here is *many-body Lyapunov spectrum*. It is not a Lieb–Robinson velocity or spectrum: Lieb–Robinson bounds constrain how rapidly information can propagate in space, whereas Lyapunov exponents describe the exponential growth or decay of singular values of a long operator product.

The answer is particularly simple for a Gaussian circuit. Evolve the maximally mixed state along the resolved record, diagonalize the resulting one-particle covariance, and retain the cumulative Born log probability of that same record. The covariance occupations determine every *relative* many-body singular value; the scalar Born weight supplies their common absolute normalization. Thus one Gaussian trajectory contains the complete many-body singular spectrum of the corresponding resolved mEO.

Throughout this note we follow the notation of Zabalo *et al.*: $\sigma_i^m(t)$ is a finite-time singular value and λ_i^m is its squared-singular-value growth rate,

$$[\sigma_i^m(t)]^2 = \exp[t\lambda_i^m + o(t)]. \quad (1)$$

This convention prevents an otherwise common factor-of-two ambiguity. In informal discussion the word “spectrum” may refer to either σ_i or λ_i ; below the notation always distinguishes them.

I. QUESTION AND SHORT ANSWER

For a fixed measurement record $\mathbf{m}$, the monitored circuit defines one resolved many-body evolution operator

II. RESOLVED TRAJECTORY AND MAXIMALLY MIXED PROBE

Let $N = 2N_x N_y$ be the number of complex fermion modes, including the two orbitals per unit cell. A resolved

record through circuit depth t is $\mathbf{m} = (m_1, \dots, m_t)$, including every measurement outcome needed to determine the feedback operations. Its chronological operator is

$$\hat{V}_{\mathbf{m}}(t) = \hat{K}_{m_t} \cdots \hat{K}_{m_2} \hat{K}_{m_1}. \quad (2)$$

No sum over records is taken. Any nonnegative positive-operator-valued-measure (POVM) weight is understood to be absorbed into the corresponding resolved Kraus operator $\hat{K}_m$.

Use the full-Fock-space maximally mixed input

$$\hat{\rho}_* = \frac{\hat{\mathbf{1}}}{2^N}. \quad (3)$$

The probability of obtaining $\mathbf{m}$ from this input is

$$p_{\mathbf{m}}(t) = \text{Tr} \left[\hat{V}_{\mathbf{m}}(t) \hat{\rho}_* \hat{V}_{\mathbf{m}}^\dagger(t) \right] = \frac{Z_{\mathbf{m}}(t)}{2^N}, \quad (4)$$

where

$$Z_{\mathbf{m}}(t) = \text{Tr} \left[\hat{V}_{\mathbf{m}}(t) \hat{V}_{\mathbf{m}}^\dagger(t) \right]. \quad (5)$$

The normalized conditional output is therefore

$$\hat{\rho}_{R,\mathbf{m}}^V(t) = \frac{\hat{V}_{\mathbf{m}}(t) \hat{V}_{\mathbf{m}}^\dagger(t)}{Z_{\mathbf{m}}(t)}. \quad (6)$$

This is the right marginal of the normalized trajectory Choi state. The maximally mixed input is consequently a spectroscopic device, not a claim about the physical state prepared in the experiment.

Equation (4) also gives the scalar information that normalized covariance dynamics discards. If the simulator accumulates

$$\omega_{\mathbf{m}}(t) = \log p_{\mathbf{m}}(t) = \sum_{u \leq t} \log p(m_u | m_1, \dots, m_{u-1}), \quad (7)$$

then

$$\boxed{\log Z_{\mathbf{m}}(t) = N \log 2 + \omega_{\mathbf{m}}(t)}. \quad (8)$$

The same realized record must supply both the covariance and $\omega_{\mathbf{m}}$.

III. EXACT GAUSSIAN RECONSTRUCTION

For the number-conserving right state in Eq. (6), define the normal covariance

$$G_{ij}^{\mathbf{m}}(t) = \text{Tr} \left[\hat{\rho}_{R,\mathbf{m}}^V(t) \hat{c}_i^\dagger \hat{c}_j \right]. \quad (9)$$

Let $\nu_a^{\mathbf{m}}(t) \in [0, 1]$, $a = 1, \dots, N$, be its occupation eigenvalues. In the corresponding natural-orbital basis the Gaussian state factorizes, so a binary occupation string $\mathbf{n} = (n_1, \dots, n_N)$, with each $n_a = 0$ or 1, has normalized many-body weight

$$w_{\mathbf{n}}^{\mathbf{m}}(t) = \prod_{a=1}^N [\nu_a^{\mathbf{m}}(t)]^{n_a} [1 - \nu_a^{\mathbf{m}}(t)]^{1-n_a}. \quad (10)$$

Since these are the eigenvalues of Eq. (6), the squared singular values and singular values of the resolved mEO are

$$[\sigma_{\mathbf{n}}^{\mathbf{m}}(t)]^2 = Z_{\mathbf{m}}(t) w_{\mathbf{n}}^{\mathbf{m}}(t), \quad (11)$$

$$\sigma_{\mathbf{n}}^{\mathbf{m}}(t) = \sqrt{Z_{\mathbf{m}}(t)} \prod_{a=1}^N [\nu_a^{\mathbf{m}}(t)]^{n_a/2} [1 - \nu_a^{\mathbf{m}}(t)]^{(1-n_a)/2}. \quad (12)$$

Equations (8) and (12) define the extraction used below. They remain valid for a nonzero exact projective branch of definite U(1) charge because $\hat{V}_{\mathbf{m}} \hat{V}_{\mathbf{m}}^\dagger$ is neutral, even when $\hat{V}_{\mathbf{m}}$ itself changes charge.

A. What the calculation actually does

For each independent sample, the numerical operation is:

1. initialize the one-particle correlation matrix as $G(0) = \frac{1}{2} \mathbf{1}_N$;
2. generate one complete record with the Born rule while applying its adaptive feedback, and accumulate the realized $\omega_{\mathbf{m}}(t)$ in Eq. (7);
3. at the chosen cycle checkpoints, diagonalize $G^{\mathbf{m}}(t)$ to obtain $\nu_a^{\mathbf{m}}(t)$ and the natural orbitals;
4. recover $\log Z_{\mathbf{m}}(t)$ from Eq. (8);
5. construct the leading ordered $\ell_i^{\mathbf{m}}(t) = \log[\sigma_i^{\mathbf{m}}(t)]^2$ using Eq. (15) and the smallest flip-cost sums; and
6. fit the checkpoint dependence to obtain one set of trajectory Lyapunov rates, then average those rates over independent records and fit their N_y dependence.

No exponentially large many-body matrix is stored. A record generated from a different initial state may still be replayed to reconstruct the spectrum of that particular $\hat{V}_{\mathbf{m}}$, but such records are sampled from the wrong measure for the simple Born average in Eq. (19). The completed ensemble therefore generated the records from the maximally mixed input itself.

B. Ordering the leading levels without exponential work

We never enumerate all 2^N binary strings. For every natural mode define its preferred and suppressed weights

$$q_a = \max(\nu_a, 1 - \nu_a), \quad r_a = \min(\nu_a, 1 - \nu_a), \quad (13)$$

and its positive flip cost

$$d_a = \log \frac{q_a}{r_a} = \left| \log \frac{\nu_a}{1 - \nu_a} \right|. \quad (14)$$

The largest many-body weight selects the preferred choice in every mode:

$$\log[\sigma_0^m(t)]^2 = \log Z_m(t) + \sum_a \log q_a. \quad (15)$$

The next levels are obtained by flipping one or more natural-mode choices. Every flip subtracts its d_a from Eq. (15). Sorting the smallest sums of the d_a 's therefore produces the leading M many-body levels with a standard heap algorithm whose cost is polynomial in N and M . In the production analysis we initially need only $M \sim 16\text{--}64$, not the complete exponentially long list.

If $\nu_a = 0$ or 1 exactly, that mode is a sharp empty or occupied Choi cap. Its preferred factor is one and its suppressed factor is zero. Flipping it produces a genuine zero singular value and a Lyapunov exponent $-\infty$. Exact caps must not be turned into finite gaps by numerical clipping. If a fixed-particle-number block of the mEO is desired instead of the full Fock-space spectrum, Eq. (10) is restricted to binary strings with the required particle number; the lowest changes are then particle-hole replacements.

C. Related tangent quantities are not substitutes

The identity-input covariance half-Jacobian has one-leg singular values

$$j_a = 2\sqrt{\nu_a(1-\nu_a)}. \quad (16)$$

These bounded numbers contain the same unoriented radial information, but the full covariance Jacobian has pairwise singular values $j_a j_b$; those are not the many-body singular values in Eq. (12). Likewise, a tangent calculation about a pure Slater trajectory sees only the coherent one-particle-one-hole sector. The maximally mixed construction is the appropriate object when the target is the complete normalized many-body spectrum of the resolved operator.

IV. LYAPUNOV EXPONENTS AND RECORD FREE ENERGY

At several depths, compute the ordered logarithms

$$\ell_i^m(t) = \log[\sigma_i^m(t)]^2. \quad (17)$$

For each independent trajectory, fit

$$\ell_i^m(t) = b_i^m + t\lambda_i^m \quad (18)$$

inside a stationary window. The intercept absorbs the fixed $N \log 2$ term and finite-depth preparation effects. The Born-ensemble exponent is

$$\lambda_i(L) = \mathbb{E}_m[\lambda_i^m(L)]. \quad (19)$$

Records generated with the simulator's Born rule are already sampled from the required distribution. Their trajectory estimates receive an ordinary arithmetic mean; they are not multiplied by a second Born weight.

The Shannon entropy of the measurement record is

$$F(L, t) = - \sum_m p_m(t) \log p_m(t) = \mathbb{E}_m[-\omega_m(t)]. \quad (20)$$

At long depth, Zabalo *et al.* identify

$$\frac{F(L, t)}{t} \longrightarrow -\lambda_0(L). \quad (21)$$

At finite depth our Gaussian reconstruction supplies the more exact leading quantity through Eq. (15). Saving the realized ω_m , rather than only the sum of binary entropies of possible outcomes, also permits recordwise covariance between free energy and spectral gaps.

The positive finite-time gap from the leading level is especially transparent:

$$\begin{aligned} \ell_0^m(t) - \ell_i^m(t) \\ = \text{the corresponding ordered sum of } d_a(t). \end{aligned} \quad (22)$$

The normalization Z_m cancels from this difference, but it remains indispensable for λ_0 , F , and the effective-central-charge analysis.

V. FOLLOWING ZABALO EQS. (3) AND (4)

For a one-dimensional critical transfer problem on a cylinder of circumference L , Zabalo *et al.* define

$$f_i(L) = -\frac{\lambda_i(L)}{\alpha L}. \quad (23)$$

Their leading finite-size forms are

$$f_0(L) = f_\infty - \frac{\pi c_{\text{eff}}}{6L^2} + \dots, \quad (24)$$

$$f_i(L) - f_0(L) = \frac{2\pi x_i^{\text{typ}}}{L^2} + \dots. \quad (25)$$

The completed analysis directly tests Eqs. (24) and (25). Ratios of gaps are not a replacement for this analysis and are not part of the primary extraction. They are retained only when their underlying finite-size coefficients pass independently.

A. What is and is not required about α

No anisotropy is needed to run the circuit, reconstruct σ_i , fit λ_i , or estimate the record free energy. It enters only when a decay rate measured per circuit cycle is converted into the conventional absolute CFT normalization in Eqs. (24) and (25). Thus α need not be known *a priori*; all raw spectral data can be collected first.

There is nevertheless no mathematical justification for silently deleting α while claiming an independent absolute c_{eff} or x_i . If we work directly in the fixed cycle convention, define

$$\tilde{f}_i(L) = -\frac{\lambda_i(L)}{L} = \alpha f_i(L). \quad (26)$$

Then the leading $1/L^2$ coefficient measures αc_{eff} , and the excited coefficients measure αx_i^{typ} .

There is also an exact way to remove α without abandoning Zabalo's finite-size relations. Let A_0 be the fitted coefficient of $1/L^2$ in $\tilde{f}_0$, and let A_i be that coefficient in $\tilde{f}_i - \tilde{f}_0$. Equations (24) and (25) imply

$$\frac{x_i^{\text{typ}}}{c_{\text{eff}}} = -\frac{A_i}{12A_0}. \quad (27)$$

This is a ratio of independently extrapolated finite-size coefficients, not a ratio of raw level gaps at one size.

For the present project this is enough. The spatial entanglement calculation already estimates the wall central charge without any time-to-space conversion. The legacy atlas finds a two-wall coefficient close to $c = 1$, together with the expected algebraic wall correlator. Combining that spatial value with Eq. (27) gives x_i^{typ} without ever specifying α . The Lyapunov free-energy fit then tests the expected finite-size form but does not provide an independent absolute value of c_{eff} . If such an independent Lyapunov-only value is required, a separate correlation or aspect-ratio determination of α must be added later.

VI. WHAT BOUNDARY CRITICALITY CHANGES

The stochastic circuit is two-dimensional, but its soft critical sector is confined to interfaces running along the periodic y direction. The effective conformal cylinder therefore uses

$$L = N_y, \quad (28)$$

$$N_x = \text{fixed}, \quad N_x \gg \xi_{\text{wall}}. \quad (29)$$

Gapped bulk degrees of freedom contribute to the nonuniversal term f_∞ . At fixed N_x , their finite- N_y corrections should be exponential rather than the wall's universal $1/N_y^2$ term. A matched uniform control is useful for diagnosing or subtracting residual bulk contributions, but it does not alter the reconstruction formulas.

The periodic domain-wall geometry contains two interfaces with opposite chirality. The atlas's full-strip entropy convention counts their combined critical content, while a contour isolating one interface carries half the logarithmic coefficient. The mEO fit must use the same convention. If the full operator contains two decoupled wall sectors, the low spectrum should show the corresponding paired structure and Eq. (24) returns the combined two-wall c_{eff} . Any division into a per-wall value requires an explicit localization and wall-decoupling check.

The full spectrum also contains many exactly capped directions and strongly gapped bulk modes. To identify the CFT levels rather than merely list ordered numbers, the natural orbitals associated with the smallest finite flip costs should be classified by

1. transverse weight near each wall;
2. momentum along y , when translation resolution is available;
3. U(1) charge sector and particle-hole content; and
4. symmetric or antisymmetric combinations of the two walls.

The occupation eigenvalues alone determine the ordered spectrum, but their eigenvectors or symmetry-resolved blocks are needed to name the corresponding operators and organize conformal towers.

VII. COMPLETED V2 CAMPAIGN AND ANALYSIS

The completed campaign uses

$$N_x = 20, \quad (30)$$

$$N_y = 20, 22, 24, 26, 28, 30, 36, 40, \quad (31)$$

$$T = 2N_y, \quad S = 100 \text{ per } N_y. \quad (32)$$

Every trajectory has the hard support-truncated domain wall at $x = 5, 15$, $n_{\text{shell}} = 1$, $\alpha_1 = 1$, $\alpha_2 = 30$, maximally mixed initialization, raster- y active-slab measurements, and perfect correction without postselection. The executed A100 code used the canonical complex128 GPU entry point `classA_U1FGTN-gpu.run_markov.circuit`. The 800 independent Born records are grouped into 160 tasks of five trajectories.

The observer saved $\omega_{\mathbf{m}}(t)$ at every cycle and diagonalized the active correlation matrix at

$$\{0, 4, 8, \dots, 2N_y\} \cup \{N_y, 3N_y/2, 2N_y\}. \quad (33)$$

The active transfer space contains $N_{\text{eff}} = 22N_y$ modes. The saved convention is

$$\ell_i(t) = \log[\sigma_i(t)]^2, \quad (34)$$

$$d_a(t) = \left| \log \frac{\nu_a(t)}{1 - \nu_a(t)} \right|, \quad (35)$$

$$\log Z_{\mathbf{m}}(t) = N_{\text{eff}} \log 2 + \omega_{\mathbf{m}}(t). \quad (36)$$

Exact $\nu_a = 0, 1$ caps retain infinite flip cost. We reconstructed all 64 saved leading levels from the occupations and $\log Z$; exact-cap-induced $-\infty$ padding is excluded from time fits rather than converted into finite levels.

A. Historical provenance and numerical validation

The local data import contains 160 NPZ files and 160 completion JSON files. We rehashed all 320 files, checked every task/configuration identity and all 800 global sample indices, and reproduced the download-manifest inventory digest. Historical products are verified against the executed source hashes pinned in the import manifest. A later repository update changed the current GPU engine hash, which is recorded separately and does not invalidate the immutable v2 products. The largest observed Hermiticity, eigensolver, eigenvector-Gram, occupation-bound, active-exterior coupling, and exterior-product residuals are respectively 0, 3.82×10^{-15} , 1.47×10^{-14} , 2.40×10^{-11} , 0, and 0. All lie below their locked tolerances.

B. Time-window and record-weight tests

Each trajectory is fit separately in the middle and late windows $[N_y, 3N_y/2]$ and $[3N_y/2, 2N_y]$; only then are the slopes averaged. Uncertainties are paired, stratified, whole-trajectory bootstrap intervals with 2000 replicates. The criterion requires the shift interval to contain zero and the magnitude to be below 10% for λ_0 and the first four gaps. Every circumference fails at least one of these five tests. For example, the failed metric sets are $\{\lambda_0, \Delta_3\}$ at $N_y = 20$, $\{\lambda_0, \Delta_3\}$ at $N_y = 30$, and $\{\Delta_1, \Delta_2, \Delta_3\}$ at $N_y = 40$. Thus $T = 2N_y$ is not accepted as a stationary Lyapunov window.

The record check must compare like with like. We therefore compare the late-window slope $d\omega/dt$ with λ_0 , rather than comparing the endpoint ratio $-\omega(2N_y)/(2N_y)$ with a window slope. The relative discrepancy is below 0.30% at every size (and below 0.15% at seven of eight sizes). This is strong closure of the reconstructed normalization. The endpoint ratio is retained only as a finite-depth diagnostic because its extensive intercept has not decayed by $2N_y$.

C. Boundary localization

Individual natural eigenvectors may rotate within nearly degenerate subspaces, so we do not attach stable wall labels to them. Instead we average the projector weight of the first four soft modes in fixed two-column windows around both walls. The combined fraction ranges from 0.9583 to 0.9688; all eight 95% bootstrap lower bounds exceed 0.947. This validates a boundary-localized soft subspace without assigning momentum, charge, parity, or conformal-operator names.

VIII. FINITE-SIZE DIAGNOSTIC AND CLAIM GATES

The all-size fit of Eq. (26) gives

$$A_0 = +1.507, qquad A_0^{95\%} = [-19.97, 24.87]. \quad (37)$$

The sign is opposite to that required for positive αc_{eff} , the interval contains zero, and the regression has $R^2 = 0.0026$. More importantly, the coefficient is not stable: adding $1/N_y^4$ changes it to 55.50, while successive $N_{y,\text{min}}$ fits change both its magnitude and sign. The maximum sensitivity is about 35.8 times the all-size coefficient, far above the preregistered 20% bound. The formal diagnostic value $\alpha c_{\text{eff}} = -2.88$ with 95% interval $[-47.49, 38.14]$ is therefore rejected and not reported as a physical estimate.

Zero-intercept fits of the first four ordered gap densities return positive raw coefficients, but every gap fails the all-size temporal gate; the first also fails finite-size stability. Consequently no αx_i is released as a provisional scaling coefficient. Because A_0 fails, the ratios $x_i/c_{\text{eff}} = -A_i/(12A_0)$ are also suppressed. The data lack momentum, charge-sector, and two-wall symmetry labels, so no conformal operator names are assigned regardless of fit quality.

Random without-replacement subsets at $S = 25, 50, 75$ and the full $S = 100$ ensemble show that sampling noise alone does not cure the coefficient instability. Increasing only the number of records at the same depth would therefore be a poor first response. The depth gate should be repaired before a larger precision ensemble is considered.

IX. INTERPRETATION AND NEXT EXPERIMENT

The v2 result separates three questions. First, the Gaussian reconstruction is numerically and probabilistically consistent: occupations and record weights reproduce the saved many-body levels and the leading late-window rate. Second, the low-cost sector is strongly localized at the two interfaces. Third, the present depth is not long enough for a controlled Zabalo-style finite-size coefficient. The latter failure is a convergence limitation, not evidence against the already established spatial boundary criticality.

The economical next experiment is a fresh $T = 4N_y$, $S = 100$ depth study at $N_y = 20, 30, 40$. It should repeat the same per-trajectory middle/late analysis at the longer depth before adding sizes or samples. This note recommends that campaign but does not launch it. The v2 outputs do not contain final covariance and random-number-generator state, so the extension cannot continue the existing trajectories and must rerun from $t = 0$.

X. CONCLUSION

A maximally mixed Gaussian trajectory gives $\hat{V}_m \hat{V}_m^\dagger / Z_m$ directly. In the completed v2 campaign, its oc-

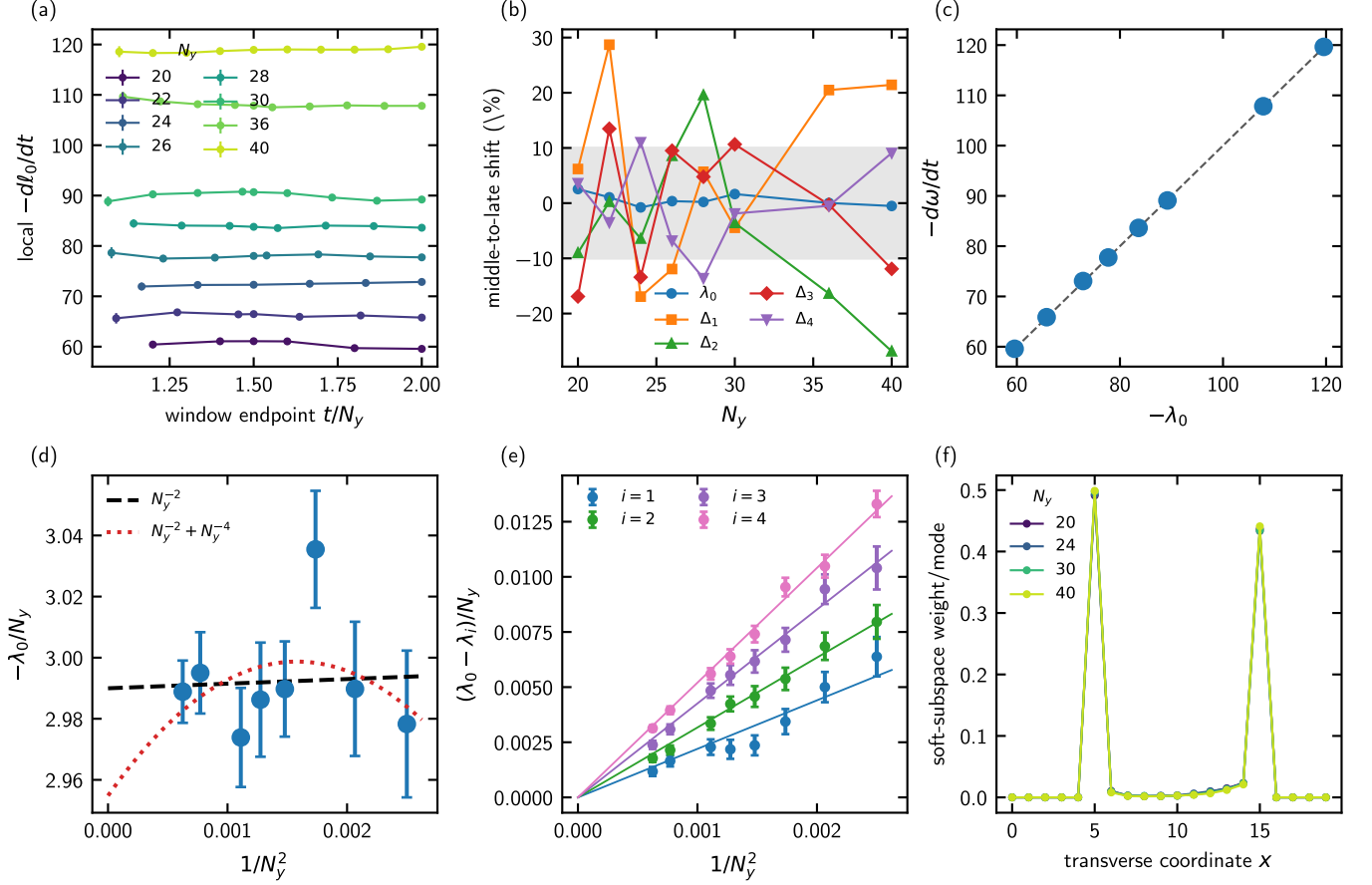


FIG. 1. Completed v2 many-body Lyapunov diagnostic. The ensemble contains $S = 100$ independent Born trajectories at each N_y , maximally mixed initialization, and total depth $T = 2N_y$. Slopes are fit trajectory by trajectory before averaging. (a) Local leading-level slopes; error bars are trajectory standard errors. (b) Middle-to-late relative shifts for λ_0 and four ordered gaps; the gray band is the 10% magnitude condition, which is necessary but not sufficient because the paired 95% bootstrap interval must also contain zero. (c) Same-late-window record-weight and leading-level slopes. (d) $-\lambda_0/N_y$ with the all-size N_y^{-2} fit and N_y^{-4} sensitivity. (e) Ordered gap densities with zero-intercept N_y^{-2} fits. (f) Aggregate first-four-mode transverse profiles; peaks occur at the two walls. Finite-size fits use $N_y = 20, 22, 24, 26, 28, 30, 36, 40$; bootstrap uncertainties use 2000 whole-trajectory replicates.

cupations reconstruct the normalized many-body singular spectrum, and the cumulative Born log weight restores the common normalization. All 800 trajectories pass provenance and numerical checks, same-window record closure is sub-percent, and the first four soft modes are 95–97% wall-localized. Nevertheless, every circumference fails at least one temporal-stability test, and the leading finite-size coefficient has the wrong sign and extreme sensitivity to fit range and subleading corrections. The campaign validates reconstruction and boundary localization; it does not establish c_{eff} , absolute operator dimensions, or a stable anisotropy-free ratio.

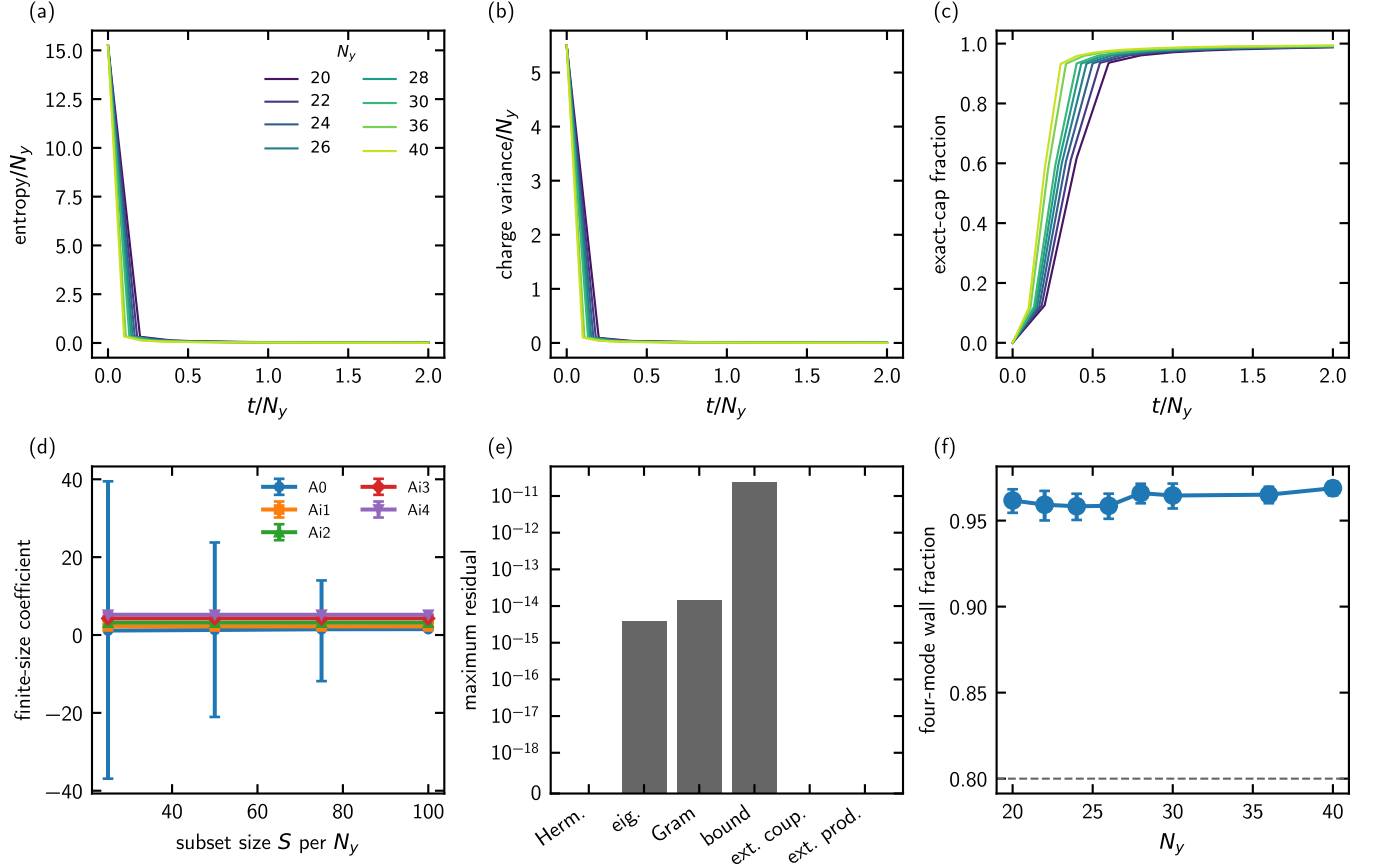


FIG. 2. Supplementary v2 diagnostics for the same $S = 100$ independent Born trajectories and $T = 2N_y$ campaign. Panels show the trajectory-averaged active entropy, charge variance, exact-cap fraction, random without-replacement sample-size sensitivity of the finite-size coefficients, maximum numerical residuals, and aggregate four-mode wall fraction. Curves use the saved estimator checkpoints; localization error bars are 95% whole-trajectory bootstrap intervals with 2000 replicates. The $S = 100$ subset is the complete ensemble and therefore has no subset-selection interval.

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